
ORIE 4154: Pricing and Market Design

Fall 2026

Linear Programming Toolkit

Primal–dual formulations, strong duality, and complementary slackness

1 What you need from linear programming

Linear programming will appear repeatedly in this course: first as a way to describe efficient allocations, and later as a way to summarize the opportunity cost of scarce resources. For our purposes, the most important ideas are not algorithms for solving LPs, but the following toolkit:

1. formulate a linear optimization problem;
2. write down its dual;
3. interpret dual variables as prices or marginal values;
4. use weak and strong duality to certify optimality;
5. use complementary slackness to connect an optimal allocation to optimal prices.

These notes follow the viewpoint in David Williamson’s ORIE 6300 Lectures 1–2: a dual solution is best understood as a systematic way to construct a bound on the primal objective. We add only the statements of strong duality and complementary slackness that we will use in ORIE 4154.

2 A linear program and its feasible region

A linear program optimizes a linear objective over a set defined by linear constraints. A useful form for this course is

$$\begin{aligned} \text{(P)} \quad & \max_{x \geq 0} \quad c^\top x \\ & \text{s.t.} \quad Ax \leq b. \end{aligned} \tag{1}$$

Here $x \in \mathbb{R}^n$ is the vector of *decision variables*. A vector x is *feasible* if it satisfies all the constraints. An optimal solution x^* is a feasible solution with the largest objective value $c^\top x$.

The feasible set is a polyhedron. In two dimensions we can often draw it, but the algebraic viewpoint is more useful: how can we prove that a candidate feasible solution is optimal without enumerating every other feasible solution?

3 Duality as a way to build upper bounds

Consider the small product-mix LP

$$\begin{aligned} \max_{x_1, x_2 \geq 0} \quad & 7x_1 + 4x_2 \\ \text{s.t.} \quad & 3x_1 + x_2 \leq 7, \\ & 2x_1 + 4x_2 \leq 8. \end{aligned} \tag{2}$$

Think of the two inequalities as two scarce resources. Any nonnegative combination of valid upper-bound constraints is still a valid upper bound. For example, multiply the first constraint by 2 and the second by 1/2:

$$2(3x_1 + x_2) + \frac{1}{2}(2x_1 + 4x_2) \leq 2(7) + \frac{1}{2}(8).$$

The left-hand side is exactly the objective, so every feasible solution satisfies

$$7x_1 + 4x_2 \leq 18.$$

The feasible point $x = (2, 1)$ attains value 18, so it must be optimal.

This example contains the idea of the dual. Let $y_i \geq 0$ be the multiplier placed on primal constraint i . We want the weighted constraints to dominate the primal objective. For the general primal (1), this means

$$A^\top y \geq c, \quad y \geq 0.$$

Then for every feasible $x \geq 0$,

$$c^\top x \leq (A^\top y)^\top x = y^\top Ax \leq y^\top b.$$

Thus every such y produces an upper bound $b^\top y$ on the primal objective. The *best* upper bound obtained in this way is another LP:

$$\begin{aligned} \text{(D)} \quad \min_{y \geq 0} \quad & b^\top y \\ \text{s.t.} \quad & A^\top y \geq c. \end{aligned} \tag{3}$$

We call (1) the *primal* and (3) its *dual*.

Theorem 3.1 (Weak duality). *For every primal-feasible x and dual-feasible y ,*

$$c^\top x \leq b^\top y.$$

Weak duality is often already enough to prove optimality: if you find feasible x and y with

$$c^\top x = b^\top y,$$

then no primal solution can do better than x and no dual solution can do better than y .

For the product-mix example, the dual is

$$\begin{aligned} \min_{y_1, y_2 \geq 0} \quad & 7y_1 + 8y_2 \\ \text{s.t.} \quad & 3y_1 + 2y_2 \geq 7, \\ & y_1 + 4y_2 \geq 4. \end{aligned} \tag{4}$$

The choice $y = (2, 1/2)$ is dual feasible and has value 18, matching the primal value at $x = (2, 1)$.

4 How to remember the dual

For the canonical pair

$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$	$\Longleftrightarrow$ $\Longleftrightarrow$ $\Longleftrightarrow$	$\begin{aligned} \min \quad & b^\top y \\ \text{s.t.} \quad & A^\top y \geq c \\ & y \geq 0, \end{aligned}$
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the mnemonic is:

Primal feature	Dual feature	Why?
one primal constraint	one dual variable	multiplier on that constraint
one primal variable	one dual constraint	coefficient of that variable in the bound
max	min	dual constructs an upper bound
$Ax \leq b$	$y \geq 0$	nonnegative multipliers preserve $\leq$
$x \geq 0$	$A^\top y \geq c$	weighted constraints must dominate objective coefficients

Williamson's Lecture 2 explains how the signs change in other standard forms. The rules most useful for us are summarized below for a *maximization* primal.

Primal	Corresponding dual condition
constraint $a_i^\top x \leq b_i$	dual variable $y_i \geq 0$
constraint $a_i^\top x = b_i$	dual variable y_i unrestricted in sign
variable $x_j \geq 0$	dual constraint $(A^\top y)_j \geq c_j$
variable x_j unrestricted in sign	dual constraint $(A^\top y)_j = c_j$

A useful consistency check is that *taking the dual twice returns the primal* (after putting both problems in compatible standard forms).

5 Strong duality

Weak duality says the dual objective is always an upper bound on the primal objective. The fundamental theorem of linear programming says that, under the usual finite-optimum conditions, there is no gap between the best primal value and the best dual bound.

Theorem 5.1 (Strong duality – course version). *If the primal LP is feasible and has a finite optimal value, then the dual also has an optimal solution and*

$$\max\{c^\top x : Ax \leq b, x \geq 0\} = \min\{b^\top y : A^\top y \geq c, y \geq 0\}.$$

The converse holds as well.

We will use strong duality as a theorem; we will not prove it in this course.

The economic interpretation is important. The primal asks for the best allocation of activities. The dual asks for prices on scarce constraints that are high enough to account for the value generated by every activity. Strong duality says that at optimum these two descriptions agree exactly.

6 Complementary slackness

Strong duality tells us that the optimal objective values agree. *Complementary slackness* tells us how an optimal primal allocation and optimal dual prices fit together.

For the canonical pair (1)–(3), define the primal slack of constraint i by

$$s_i = b_i - a_i^\top x \geq 0,$$

and the dual slack of variable j by

$$t_j = (A^\top y)_j - c_j \geq 0.$$

The duality gap separates into these two kinds of slack:

$$\begin{aligned} b^\top y - c^\top x &= y^\top (b - Ax) + x^\top (A^\top y - c) \\ &= \sum_i y_i s_i + \sum_j x_j t_j. \end{aligned} \tag{5}$$

Every term on the last line is nonnegative for feasible x and y . Thus the gap is zero exactly when every product $y_i s_i$ and $x_j t_j$ is zero. This is the source of complementary slackness.

Theorem 6.1 (Complementary slackness). *A primal-feasible x and dual-feasible y are both optimal if and only if*

$$y_i (b_i - a_i^\top x) = 0 \quad \text{for every primal constraint } i, \tag{6}$$

$$x_j ((A^\top y)_j - c_j) = 0 \quad \text{for every primal variable } j. \tag{7}$$

The equations are easier to remember in words.

Resource side: if a dual price y_i is positive, the corresponding primal resource constraint must be tight. Equivalently, if a resource is left slack, its dual price must be zero.

Activity side: if a primal activity x_j is used positively, its dual constraint must be tight. Equivalently, a positive activity must exactly cover its value at the prevailing dual prices.

For the product-mix example, $x = (2, 1)$ makes both resource constraints tight, and $y = (2, 1/2)$ has both dual prices positive. Since both x_1, x_2 are positive, both dual inequalities are also tight:

$$3(2) + 2\left(\frac{1}{2}\right) = 7, \quad 2 + 4\left(\frac{1}{2}\right) = 4.$$

Complementary slackness therefore certifies that this primal allocation and these dual prices belong together.

The zero-price implication becomes clearer if we add a redundant resource constraint $x_1 + x_2 \leq 10$. It is slack at $x = (2, 1)$, so complementary slackness forces its dual price to be zero. Conversely, $x_j = 0$ imposes no tightness condition: an unused activity may cost strictly more than its value at the dual resource prices.

6.1 Why complementary slackness is economically useful

In market-design and revenue-management applications, dual variables often have a price interpretation. Complementary slackness then becomes an economic equilibrium condition:

- a resource commands a positive price only when it is scarce at the optimum;
- an activity is chosen only when its value is high enough to cover the prices of the resources it uses;
- unused/slack resources can have zero marginal value;
- chosen activities are exactly competitive at the optimal resource prices.

This is why duality will recur in ORIE 4154. In the assignment-market example, dual variables become item prices and buyer utilities. In network revenue management, they become bid prices for capacity.

7 A short checklist

Given a new LP, you should be able to do the following.

1. **Write the primal clearly.** Identify variables, objective, and constraints.
2. **Write the dual.** One dual variable per primal constraint; one dual constraint per primal variable.
3. **Check weak duality.** Ask: does a dual-feasible solution really give the correct type of bound on the primal?
4. **Use strong duality.** Matching primal and dual objective values certify optimality.
5. **Use complementary slackness.** Positive primal variables force tight dual constraints; positive dual variables force tight primal constraints.
6. **Interpret the dual variables.** In this course they will often be prices, marginal values, or opportunity costs.

Recommended review

These notes are based closely on the first two lectures of David Williamson's ORIE 6300 notes:

- [Lecture 1: introduction to LPs and duality](#). The central idea is that nonnegative linear combinations of constraints generate objective-value bounds, and the best such bound is the dual LP.
- [Lecture 2: standard forms and duality](#). Use this if you need to review how equality constraints, nonnegative/free variables, and different primal forms change the dual.

For ORIE 4154, you should know the statements and uses of weak duality, strong duality, and complementary slackness. You do not need the proof of strong duality or Farkas' lemma.